

Notes: Holmström & Tirole (JPE, 1998)

Private and Public Supply of Liquidity

Contents

1	Big picture	3
2	Model environment and notation	3
2.1	Three-date timeline	3
2.2	Technology and moral hazard	4
2.3	First-best and pledgeable returns	4
2.4	Positive-NPV and no-self-financing assumptions	4
3	Partial-equilibrium contract: the firm's demand for liquidity	5
3.1	The second-best contract	5
3.2	Why the entrepreneur receives the minimum stake	5
3.3	Equity multiplier and continuation cutoff	5
3.4	Implementation: credit line or liquid reserve	6
3.5	Proposition 1	6
4	Endogenous private liquidity without aggregate uncertainty	6
4.1	Private assets only	6
4.2	Why anonymous markets can fail	7
4.3	Intermediation and credit lines	7
4.4	Proposition 2	7
5	Aggregate uncertainty and public liquidity	8
5.1	Why private pooling breaks down	8
5.2	Government bonds at par	8
5.3	Proposition 3	8
5.4	Liquidity premia and free-riding	8
5.5	Private-sector policy with costly bonds	9
5.6	Proposition 4	9
5.7	State-contingent bonds and active liquidity policy	9
6	Theoretical design: assumptions that drive the results	10
6.1	Limited pledgeability	10
6.2	No unsecured consumer borrowing	10
6.3	Government enforcement power	10
6.4	Idiosyncratic versus aggregate shocks	10
6.5	What the paper does not fully prove	10
7	Tables and figures	11

7.1	Figure 1: Timing of the liquidity problem	11
7.2	Compact map of the propositions	12
8	Short summary	12
9	Conclusion	13
9.1	Core conclusion	13
9.2	Economic intuition	13
9.3	Takeaway	13

1 Big picture

- **Question.** Why do firms demand liquidity, when can private financial markets supply it, and when is there a role for government-supplied liquidity?
- **Core idea.** Firms demand liquidity because moral hazard limits pledgeable income. A firm may have a positive-continuation-value project at date 1 but be unable to raise enough outside funds to cover a liquidity shock.
- **Liquidity definition.** Liquidity is the availability of instruments that transfer purchasing power across dates and can be used to cover future financing needs.
- **Model environment.** Firms invest at date 0, receive a random liquidity shock at date 1, and realize returns at date 2 if the project continues. The entrepreneur must retain enough upside to choose high effort.
- **First main result.** The optimal contract sets an initial investment level and a date-1 continuation cutoff. It can be implemented either by a credit line or by liquid reserves with a covenant.
- **Private markets without aggregate uncertainty.** If liquidity shocks are idiosyncratic, the private sector has enough aggregate liquidity, but anonymous markets can misallocate it. Intermediaries improve on markets by pooling liquidity risk through credit lines.
- **Government without aggregate uncertainty.** With independent shocks, government securities have no special liquidity role. Private intermediation can implement the productive optimum.
- **Aggregate uncertainty.** If all firms face the same liquidity shock, private pooling fails. The productive sector may need government securities because no firm has excess liquidity when the aggregate shock is bad.
- **Liquidity premium.** Government bonds can sell above par because firms value them as productive liquidity inputs. This can coexist with risk-neutral consumers who do not value liquidity.
- **Active liquidity management.** Noncontingent government bonds can induce inefficient partial liquidation when bonds are costly. State-contingent bonds, interpreted as active monetary or liquidity policy, supply liquidity only in high-need states.
- **Economic takeaway.** The government may need to supply aggregate liquidity not because private agents are irrational, but because private claims are constrained by pledgeability and cannot commit future consumer endowments.

2 Model environment and notation

2.1 Three-date timeline

- At date 0, a financial contract is signed and the firm invests I .
- At date 1, the firm receives a liquidity shock ρI .
- If the shock is paid, the project continues. If not, the project is abandoned and yields zero.

- At date 2, after continuation, the entrepreneur chooses effort; the project succeeds with probability p_H if he behaves and p_L if he shirks.
- If the entrepreneur shirks, he obtains private benefit BI .

Interpretation: The timing is essential. The firm must decide how much liquidity protection to arrange before knowing the date-1 shock, and the moral hazard problem limits how much outside investors can be promised later.

2.2 Technology and moral hazard

The project has constant returns to scale. If it succeeds, it pays RI ; if it fails, it pays zero. The entrepreneur chooses effort privately:

$$p_H > p_L, \quad \Delta p = p_H - p_L > 0.$$

The entrepreneur's private benefit from shirking is BI . The incentive constraint requires the entrepreneur to retain a minimum success payoff:

$$R_f(\rho)\Delta p \geq B.$$

The minimum retained payoff per unit of investment is

$$R_b = \frac{B}{\Delta p}.$$

Interpretation: The moral hazard wedge is the source of limited pledgeability. Outside investors cannot be promised the full project return because the entrepreneur must keep enough upside to behave.

2.3 First-best and pledgeable returns

The first-best continuation cutoff is

$$\rho_1 = p_H R.$$

The project should continue whenever $\rho \leq \rho_1$ because expected continuation value exceeds the liquidity cost. The pledgeable date-1 return per unit of investment is

$$\rho_0 = p_H \left(R - \frac{B}{\Delta p} \right).$$

Because the entrepreneur must retain $B/\Delta p$, outsiders can claim only $\rho_0 I$, not $\rho_1 I$.

Interpretation: The inequality $\rho_0 < \rho_1$ is the key wedge. The firm may have a socially valuable continuation opportunity, yet be unable to raise enough funds from outside investors.

2.4 Positive-NPV and no-self-financing assumptions

The project is valuable if the entrepreneur behaves but not if he shirks:

$$\int \max\{p_H R - \rho, 0\} f(\rho) d\rho - 1 > 0 > \int \max\{p_L R + B - \rho, 0\} f(\rho) d\rho - 1.$$

The no-self-financing condition is

$$\int_0^{\rho_0} (\rho_0 - \rho) f(\rho) d\rho < 1.$$

Interpretation: The first condition gives a reason to finance projects. The second rules out unlimited investment by ensuring that even the best pledgeable continuation policy cannot fully finance itself.

3 Partial-equilibrium contract: the firm's demand for liquidity

3.1 The second-best contract

A contract specifies

$$C = \{I, \lambda(\rho), R_f(\rho)\},$$

where $\lambda(\rho) = 1$ means continuation and $R_f(\rho)$ is the entrepreneur's success payoff. The second-best program is

$$\max_{I, \lambda, R_f} I \int p_H R_f(\rho) \lambda(\rho) f(\rho) d\rho - A \quad (1)$$

$$\text{s.t. } I \int \{p_H [R - R_f(\rho)] - \rho\} \lambda(\rho) f(\rho) d\rho \geq I - A, \quad (2)$$

$$R_f(\rho) \Delta p \geq B \quad \text{for every } \rho. \quad (3)$$

Interpretation: The firm maximizes its expected surplus subject to investor break-even and entrepreneur incentive compatibility.

3.2 Why the entrepreneur receives the minimum stake

Since investment scale is valuable when the marginal surplus is positive, the entrepreneur sets

$$R_f(\rho) = R_b = \frac{B}{\Delta p}.$$

This maximizes the amount pledgeable to investors and therefore maximizes initial scale.

Interpretation: The entrepreneur does not voluntarily keep more than the incentive-compatible minimum because doing so reduces external financing capacity.

3.3 Equity multiplier and continuation cutoff

For a cutoff rule $\hat{\rho}$, define the marginal net social return

$$m(\hat{\rho}) = \int_0^{\hat{\rho}} (\rho_1 - \rho) f(\rho) d\rho - 1.$$

The investment capacity is

$$I = k(\hat{\rho}) A,$$

where the equity multiplier is

$$k(\hat{\rho}) = \frac{1}{1 + \int_0^{\hat{\rho}} \rho f(\rho) d\rho - F(\hat{\rho}) \rho_0}.$$

The firm's net payoff is

$$U_f(\hat{\rho}) = m(\hat{\rho}) k(\hat{\rho}) A.$$

The second-best cutoff is

$$\rho^* \in (\rho_0, \rho_1),$$

chosen to maximize $m(\hat{\rho})k(\hat{\rho})$.

Interpretation: Raising the cutoff improves continuation efficiency but requires more liquidity protection, which reduces initial scale. The optimal cutoff trades off initial investment against the probability of wasteful abandonment.

3.4 Implementation: credit line or liquid reserve

The second-best can be implemented in either of two equivalent partial-equilibrium ways:

- raise $I - A$ at date 0 and provide an irrevocable credit line of size ρ^*I ;
- raise $(1 + \rho^*)I - A$ at date 0 and impose a covenant requiring the firm to keep ρ^*I in liquid reserves.

Interpretation: A credit line and a reserve covenant are equivalent when liquid assets are exogenously available at par. They stop being equivalent in general equilibrium when liquid assets themselves are scarce and can carry a premium.

3.5 Proposition 1

- The firm invests $I = k(\rho^*)A$.
- The project continues if and only if $\rho \leq \rho^*$.
- The cutoff lies strictly between pledgeable return and social return: $\rho_0 < \rho^* < \rho_1$.
- If the project succeeds, the entrepreneur receives $(B/\Delta p)I$ and investors receive $[R - (B/\Delta p)]I$.
- The allocation can be implemented by a credit line or by reserves with a liquidity covenant.

Economic message: The firm has a genuine demand for liquidity because financial constraints can force inefficient liquidation even when continuation has positive social value.

4 Endogenous private liquidity without aggregate uncertainty

4.1 Private assets only

Now there is a continuum of firms with independent liquidity shocks and no aggregate uncertainty. There is no storage technology and no government debt. The only way to transfer wealth is through claims on firms.

The deterministic aggregate liquidity need for the productive optimum is

$$D = I \int_0^{\rho^*} \rho f(\rho) d\rho.$$

If firms issue claims and hold market claims, the date-1 value of external claims on continuing firms is

$$V_1 = F(\rho^*)\rho_0 I.$$

After reinvestment needs are paid, the market portfolio value is

$$S_1 = V_1 - D = \left[F(\rho^*)\rho_0 - \int_0^{\rho^*} \rho f(\rho) d\rho \right] I = I - A.$$

Interpretation: The private sector has enough aggregate value, but this does not mean every firm has the right amount of liquidity exactly when it needs it.

4.2 Why anonymous markets can fail

Suppose each firm holds a fraction of the market portfolio. A firm can withstand shocks satisfying

$$\rho I \leq \rho_0 I + \alpha S_1.$$

For market holdings to support the cutoff ρ^* , the paper derives the condition

$$\rho^* + \int_0^{\rho^*} \rho f(\rho) d\rho < [1 + F(\rho^*)]\rho_0.$$

If pledgeable income ρ_0 is small, this condition fails.

Interpretation: The problem is misallocation, not aggregate scarcity. Lucky firms hold liquidity they do not need; unlucky firms cannot sell enough claims to continue.

4.3 Intermediation and credit lines

An intermediary can pool idiosyncratic liquidity shocks:

- It holds claims on all firms.
- It grants each firm a credit line up to $\rho^* I$.
- Firms draw only what they need because unused liquidity has no private value to them.
- Low-shock firms effectively cross-subsidize high-shock firms through the intermediary.

Interpretation: A credit line is more state-contingent than anonymous market holdings. It delivers liquidity to firms based on realized need rather than giving every firm the same market portfolio.

4.4 Proposition 2

- Without aggregate uncertainty, anonymous financial markets cannot always implement the productive optimum.
- Private intermediaries can implement the productive optimum by providing credit lines and cross-subsidizing firms.
- Government securities have no liquidity role in this environment; if issued, they must be offered at par and do not affect real allocations.

Economic message: Private intermediation is valuable because it reallocates scarce liquidity across firms after idiosyncratic shocks are realized.

5 Aggregate uncertainty and public liquidity

5.1 Why private pooling breaks down

With pure aggregate uncertainty, all firms face the same liquidity shock. When $\rho_0 < \rho \leq \rho^*$, every firm needs more liquidity than it can raise by pledging future returns.

Interpretation: Intermediation cannot help when every firm is hit at the same time. There are no lucky firms from which to reallocate excess liquidity.

5.2 Government bonds at par

If taxation has no net social cost, the government can issue bonds at date 0. Firms or intermediaries hold bonds of size

$$(\rho^* - \rho_0)I.$$

Together with the firm's own pledgeable value $\rho_0 I$, this covers shocks up to $\rho^* I$.

Interpretation: Government bonds are valuable because the government can commit future consumer endowments through taxation, something private agents cannot do with unsecured promises.

5.3 Proposition 3

- With aggregate uncertainty, the private sector is not self-sufficient in liquidity.
- If taxation has no net cost, government bonds issued at the market rate implement the productive optimum.
- Bonds reduce date-0 productive investment because firms save more for date-1 liquidity needs.
- Bonds increase date-1 reinvestment and reduce liquidation, so expected output and firm value rise.

Economic message: Government debt can be a productive liquidity input. It crowds out some initial investment but preserves more projects when the aggregate liquidity shock is bad.

5.4 Liquidity premia and free-riding

If government securities sell at price $q > 1$ per unit of expected return, then $q - 1$ is a liquidity premium. Consumers do not value liquidity, so they do not buy premium bonds. Firms value the bonds because they insure production.

A free-riding problem appears: a firm may prefer to buy shares in firms that hold bonds instead of buying bonds itself. If all firms do this, no bonds are held and liquidity protection disappears.

Interpretation: Liquidity becomes a public-good-like input inside the productive sector when bonds carry a premium.

5.5 Private-sector policy with costly bonds

Let zI be the face value of bonds purchased and $\lambda(\rho, z)$ be the fraction of firms continued. If bonds are costly, the private sector economizes on them and uses partial liquidation.

The optimal continuation rule has two thresholds, $\hat{\rho}$ and $\bar{\rho}$:

$$\lambda(\rho, z) = \begin{cases} 1, & \rho \leq \hat{\rho}, \\ \frac{\hat{\rho} - \rho_0}{\rho - \rho_0}, & \hat{\rho} < \rho \leq \bar{\rho}, \\ 0, & \rho > \bar{\rho}. \end{cases}$$

Interpretation: Partial liquidation lets the private sector economize on bonds that are costly ex ante. But from the social point of view, this is inefficient relative to state-contingent liquidity.

5.6 Proposition 4

If $q > 1$, the private sector's optimal solution has thresholds

$$\rho_0 < \hat{\rho} < \bar{\rho} < \rho_1.$$

It buys

$$zI = (\hat{\rho} - \rho_0)I$$

in government bonds. All firms continue if $\rho \leq \hat{\rho}$; only a fraction

$$\frac{\hat{\rho} - \rho_0}{\rho - \rho_0}$$

continue if $\hat{\rho} < \rho \leq \bar{\rho}$; and no firms continue above $\bar{\rho}$.

Economic message: Costly liquidity generates socially wasteful liquidation. The private sector protects itself, but it underuses the liquidity input because it must pay a premium for bonds it may not need ex post.

5.7 State-contingent bonds and active liquidity policy

The social optimum chooses a cutoff $\rho^\#$ and an all-or-nothing continuation rule:

$$\lambda^*(\rho) = \begin{cases} 1, & \rho \leq \rho^\#, \\ 0, & \rho > \rho^\#. \end{cases}$$

where

$$\rho^\# - \rho_0 = \frac{\rho_1 - \rho_0}{\delta q}.$$

State-contingent bonds pay

$$x(\rho) = \begin{cases} 0, & \rho \leq \rho_0 \text{ or } \rho > \rho^\#, \\ \rho - \rho_0, & \rho_0 < \rho \leq \rho^\#. \end{cases}$$

Interpretation: State-contingent bonds supply liquidity only when firms need it. This avoids both ex ante overpayment for unused liquidity and ex post partial liquidation. The paper interprets this as a metaphor for active liquidity management: tighten liquidity in good aggregate states and loosen it in bad aggregate states.

6 Theoretical design: assumptions that drive the results

6.1 Limited pledgeability

The moral hazard/incentive constraint is the central friction. Without the need to leave the entrepreneur $B/\Delta p$, the pledgeable return would equal the social return and firms would not demand liquidity protection.

6.2 No unsecured consumer borrowing

Consumers cannot commit future endowments unless promises are backed by marketable assets. This prevents private agents from creating unlimited liquidity by issuing unsecured claims.

6.3 Government enforcement power

The government can tax and enforce payments. This lets it commit future consumer endowments in a way private agents cannot. That is why public debt can expand aggregate liquidity.

6.4 Idiosyncratic versus aggregate shocks

The distinction between independent shocks and common shocks is decisive:

- with idiosyncratic shocks, private liquidity pooling works;
- with aggregate shocks, private pooling fails and public liquidity can help.

6.5 What the paper does not fully prove

- The paper is theoretical and does not estimate the model.
- The liquidity shock is stylized; it can represent cost overruns, revenue shortfalls, or operating liquidity needs.
- Government securities are a metaphor for broader public liquidity management, not only literal Treasury-bond issuance.
- The model abstracts from risk aversion, banks' own agency problems, taxes beyond the deadweight-cost parameter, and many institutional details.

Interpretation: The results are sharp because the model isolates a single mechanism: limited pledgeability creates liquidity demand, and aggregate liquidity shortages justify public liquidity supply.

7 Tables and figures

The original paper contains one formal figure and no empirical tables. The figure is directly cropped from the uploaded paper. To keep the notes compact, the four main propositions are grouped in a summary table below the figure.

7.1 Figure 1: Timing of the liquidity problem

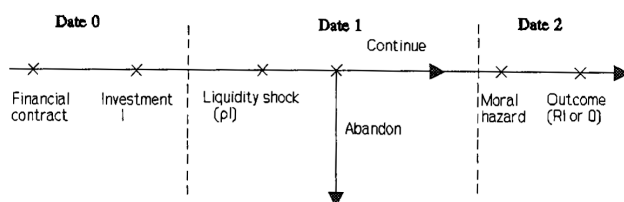


FIG. 1

Figure 1: Original Figure 1: Timing of financial contract, investment, liquidity shock, continuation or abandonment, moral hazard, and final outcome.

Read:

- At date 0, the firm signs a financial contract and chooses investment scale I .
- Before date 1, the project receives a liquidity shock pI .
- At date 1, the project is continued if financing is available; otherwise it is abandoned.
- Moral hazard occurs only after continuation, before the date-2 outcome.
- The timing explains why liquidity must be arranged ex ante: the firm may not be able to raise enough funds after the shock.

Economic message: The firm needs liquidity not because investment has negative NPV, but because future pledgeable income is too low to finance all positive-continuation-value states.

7.2 Compact map of the propositions

Result	Setting	Main message
Proposition 1	Single firm with exogenous liquidity supply	The second-best firm invests $I = k(\rho^*)A$ and continues iff $\rho \leq \rho^*$, with $\rho_0 < \rho^* < \rho_1$. It can be implemented by a credit line or reserve covenant.
Proposition 2	Continuum of firms, independent liquidity shocks	Anonymous markets can misallocate liquidity; intermediaries implement the productive optimum through credit lines. Government securities have no real role.
Proposition 3	Pure aggregate liquidity shocks, zero tax cost	Private liquidity is insufficient. Government bonds issued at the market rate implement the productive optimum and raise expected output and firm value.
Proposition 4	Aggregate shocks with costly government bonds, $q > 1$	Costly bonds induce private economizing and partial liquidation. The private optimum uses thresholds $\hat{\rho}$ and $\bar{\rho}$; state-contingent public liquidity improves on noncontingent bonds.

Economic message: The propositions move from firm-level liquidity demand to private liquidity pooling and then to public liquidity supply. The role of government appears only when aggregate liquidity risk prevents private cross-subsidization.

8 Short summary

- The paper builds a theory of liquidity demand by firms using a dynamic moral hazard model.
- Because the entrepreneur must retain a minimum stake, outside investors can be promised only a pledgeable return ρ_0 , below the social return ρ_1 .
- The firm may therefore be forced to liquidate a positive-continuation-value project after a date-1 liquidity shock.
- The optimal contract chooses initial scale and a continuation cutoff ρ^* .
- The firm can implement the second-best allocation through a credit line or by holding liquid reserves with a covenant.
- With independent liquidity shocks, the private sector has enough aggregate liquidity, but anonymous markets may misallocate it.
- Intermediaries improve on markets by pooling idiosyncratic shocks and providing credit lines.
- With aggregate liquidity shocks, private liquidity pooling fails because all firms need liquidity simultaneously.
- Government securities can help because the government can commit future consumer endowments through taxation.

- If government bonds sell at a premium, firms may free-ride on other firms' liquidity holdings and may partially liquidate to economize on bonds.
- State-contingent bonds dominate noncontingent bonds because they supply liquidity only when needed.
- The policy interpretation is that liquidity should be tightened in good aggregate states and loosened in bad aggregate states.

9 Conclusion

9.1 Core conclusion

Holmström and Tirole show that liquidity shortages arise from limited pledgeability. Firms need advance liquidity because positive-continuation-value projects may not be able to raise enough funds after a shock. When shocks are idiosyncratic, private intermediaries can pool liquidity risk. When shocks are aggregate, government-supplied liquidity can be welfare improving.

9.2 Economic intuition

The same agency problem that limits firm financing also limits the private supply of liquidity. A firm cannot pledge its full future return because the entrepreneur must be incentivized. When every firm is hit by a common shock, private claims are not enough: there is no unaffected sector that can transfer resources. Government debt works because the government can tax future endowments and thereby create claims that private agents cannot replicate.

9.3 Takeaway

The paper provides a theory of public liquidity management. Treasury securities and similar public instruments are not merely stores of value; they can be inputs into productive investment when firms face binding financing constraints. The strongest policy lesson is that the optimal public supply of liquidity should be state-contingent: provide liquidity when aggregate shocks are bad and withdraw it when liquidity demand is low.